Project Title – Airfare Price Prediction Using Machine Learning

1. Summary of Problem Statement, Data and Findings

1.1 Problem Statement

Airfare prices in the U.S. commercial aviation industry vary significantly across different airport routes due to factors such as flight distance, passenger demand, airline competition, market share, and seasonal travel patterns. These pricing strategies are driven by complex Revenue Management Systems (RMS), which operate like "black boxes," making it difficult for travelers and analysts to understand why fares fluctuate.

This project aims to develop a machine learning system that predicts the average fair market airfare for domestic routes and identifies the key drivers behind fare variations, such as the effect of monopoly power versus route distance

1.2 Data Source and Description

The dataset used is the U.S. Consumer Airfare Report (Table 1a) published by the U.S. Department of Transportation (Bureau of Transportation Statistics). It provides quarterly data on average fares, passenger volumes, and market shares for U.S. airport-pair markets.

The dataset includes route and carrier-related features such as:

- fare (Target variable: average passenger fare)
- nsmiles (Non-stop distance in miles)
- passengers (Average passengers per day)
- large_ms (Market share of the largest carrier)
- lf_ms (Market share of the low-fare carrier)
- Categorical identifiers such as airport_1, airport_2, carrier_lg, carrier_low

1.3 Key Findings from EDA

Exploratory Data Analysis provided the following insights:

- The target variable fare is slightly right-skewed, with most fares ranging between $150 and $300.

- The passengers feature is highly skewed, showing that a small number of popular routes carry the majority of traffic.
- The relationship between distance (nsmiles) and fare is positive but only moderately strong (correlation ≈ 0.58), confirming that distance alone does not fully explain airfare variation.
- Routes where a single airline holds more than 70% market share (large_ms) tend to have significantly higher fares, supporting the "monopoly premium" effect.

The following visualizations summarize the key patterns observed in the dataset:

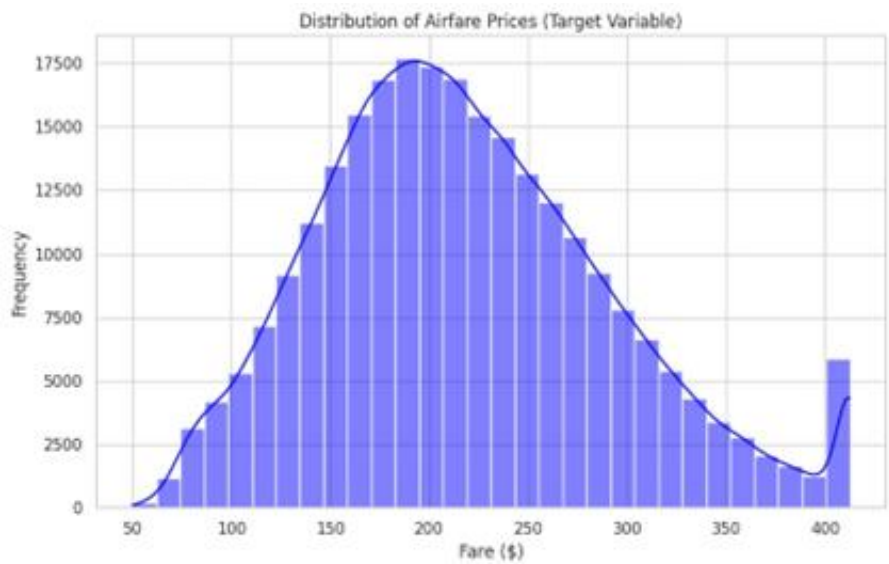


Fig 1.1: Distribution of Airfare Prices

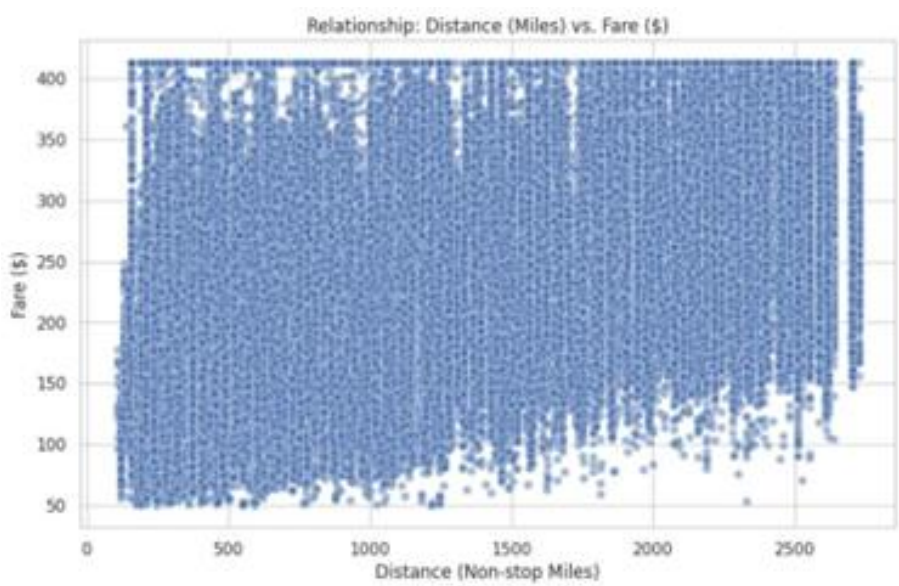


Fig 1.2: Relation between Distance and Fare

1.4 Baseline Model Findings

A Multiple Linear Regression model was trained as the baseline benchmark for airfare prediction. The baseline achieved an R^2 score of 0.58, meaning it explained 58% of the variation in airfare. This indicates airfare pricing is strongly non-linear and requires advanced models such as tree-based ensembles to capture complex interactions between competition and route characteristics.

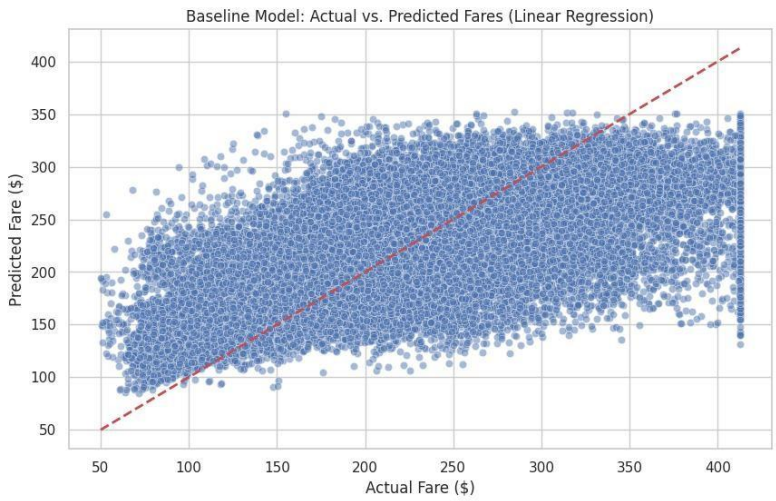


Fig 1.3: Baseline Model Performance (Actual vs Predicted – Linear Regression)

2. Overview of Final Process

2.1 Business Understanding

The U.S. commercial aviation industry uses complex Revenue Management Systems (RMS) to dynamically price airline tickets. Due to demand fluctuations, route distance, airline competition, and market concentration, fares often vary significantly even for similar routes. These pricing strategies operate like “black box” systems, creating limited transparency for travelers and making it difficult to estimate a fair market value for a given ticket.

The objective of this capstone project was to build an end-to-end machine learning system that predicts the average airfare for U.S. domestic routes and explains the key drivers influencing fare variation. In addition, the solution was deployed as a user-facing web application (“SkyCast AI”) to demonstrate real-world usability.

2.1 Problem-Solving Methodology

A complete machine learning lifecycle was followed, starting from data acquisition and cleaning, progressing through exploratory analysis and feature engineering, and culminating in advanced model training, evaluation, and deployment. The approach was iterative, where insights from EDA and early model performance directly informed feature engineering decisions and final model selection

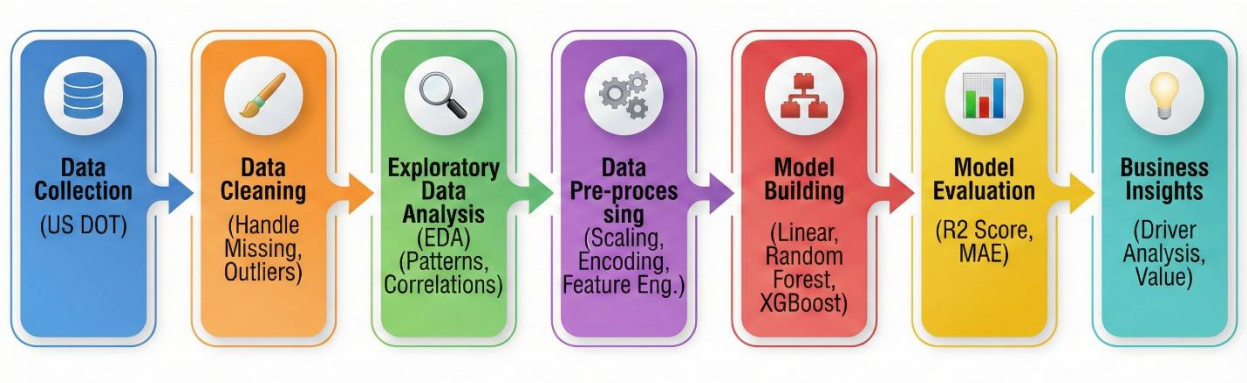


Fig 2.1: End-to-End Methodology Workflow (Data → Model → Insights)

2.3 Data Collection and Salient Features

The dataset used in this project is the U.S. Consumer Airfare Report (Table 1a) published by the U.S. Department of Transportation (Bureau of Transportation Statistics). The dataset provides quarterly route-level information for U.S. airport-pair markets, including average fare values, passenger demand, route distance, and market share indicators.

The most salient features used for airfare prediction include:

- fare (Target variable: average passenger fare)
- nsmiles (Non-stop distance in miles)
- passengers (Average passengers per day)
- large_ms (Market share of the largest carrier)
- lf_ms (Market share of the low-fare carrier)
- airport_1, airport_2, carrier_lg, carrier_low (categorical identifiers)

2.4 Data Cleaning and Preparation

To ensure the dataset was consistent and suitable for modeling, the following data preparation steps were applied:

- **Missing Value Handling:** Minor missing values were found mainly in market share-related columns. Numerical missing values were imputed using the median, while categorical missing values were imputed using the mode.

- **Redundant Column Removal:** Geocoded_City1 and Geocoded_City2 were removed as they contained redundant geographical information already represented through airport codes.

- **Data Leakage Prevention:** fare_lg and fare_low were identified as leakage variables because they are direct components of the average fare. These columns were removed to ensure the model learns from route and competition characteristics rather than copying correlated fare values.

- **Outlier Treatment:** Outliers in fare and passenger volume were treated using the Interquartile Range (IQR) method, and extreme values were capped to improve model stability.

2.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to identify patterns and validate assumptions before model training. The main analyses included:

- **Univariate Analysis:** Understanding the distribution of fare and passenger demand.

- **Bivariate Analysis**: Studying relationships such as fare vs distance and fare vs market share indicators.

- **Correlation Analysis:** Checking multicollinearity and identifying strong feature dependencies.

2.6 Data Preprocessing and Feature Engineering

To capture real-world pricing dynamics and improve model learning, preprocessing and feature engineering were applied:

- **Log Transformation:** The passengers variable was log-transformed to reduce skewness and improve regression performance.

- **Feature Creation:**

    - competition_ratio = large_ms / lf_ms to quantify dominance of the largest carrier relative to low-fare competitors.

- route_id = airport_1 + airport_2 to capture route-level identity and route-specific signatures.

- **Scaling:** StandardScaler was applied to numerical variables such as distance and passengers to ensure consistent feature ranges during model training.

EDA insights guided feature selection and helped justify the need for advanced non-linear modeling beyond linear regression.

2.7 Modeling Approach

The modeling strategy was designed to move from a benchmark model to an advanced model capable of handling non-linear pricing patterns:

- **Baseline Model:** Multiple Linear Regression was trained to establish benchmark performance. The baseline achieved an R^2 score of 0.58, confirming that airfare pricing behavior is not purely linear.

- **Final Model:** XGBoost regression was selected as the final modeling approach due to its ability to capture complex non-linear relationships and interactions between route distance, passenger demand, and market concentration.

2.8 Model Evaluation Strategy

To evaluate the success of the models, standard regression metrics were used:

- R^2 Score to measure explained variance

- MAE (Mean Absolute Error) to measure average absolute prediction error

- RMSE (Root Mean Squared Error) to penalize larger errors

- MAPE (Mean Absolute Percentage Error) to measure relative prediction error

These metrics provide a balanced evaluation of both accuracy and robustness across different route types and fare ranges.

2.9 Deployment (Frontend Application)

A frontend web application was developed using the Flask framework, with the interface designed using HTML, CSS, and JavaScript. The application allows users to input route-related details and instantly obtain predicted airfare values, demonstrating the real-world usability and accessibility of the trained machine learning model.

3. Step-by-Step Walkthrough of the Solution

3.1 Step 1: Data Collection

The project began by collecting airfare route-level data from the U.S. Consumer Airfare Report (Table 1a). This dataset was selected because it provides structured, reliable quarterly information on average fare, passenger volume, distance, and competition indicators for U.S. domestic routes. This step established the foundation for building a supervised regression model.

3.2 Step 2: Data Understanding and Feature Selection

After loading the dataset, the target variable was defined as fare. Predictor variables were selected based on business relevance:

- Distance (nsmiles) as a cost driver
- Passengers as a demand driver
- large_ms and lf_ms as competition drivers
- Airports and carriers as route identity signals

This step ensured the model was trained on meaningful factors that influence airfare pricing.

3.3 Step 3: Missing Value Handling

A missing value check was performed. Minor missing values were found in market share-related features. Median and mode imputation strategies were applied to maintain dataset completeness and avoid losing records.

This ensured a stable dataset for downstream modeling.

3.4 Step 4: Data Cleaning, Leakage Removal, and Outlier Treatment

Redundant and leakage-prone columns were removed to improve reliability:

- fare_lg and fare_low were removed to prevent leakage and unrealistic model evaluation.
- Geocoded_City1 and Geocoded_City2 were dropped as redundant fields

Outliers in fare and passenger volume were capped using the IQR method to prevent extreme routes from dominating model learning.

3.5 Step 5: Exploratory Data Analysis (EDA) and Key Insights

EDA was conducted to understand distributions and relationships:

- Fare values were slightly right-skewed, with most fares concentrated between $150 and $300.
- Passenger volume was highly skewed, indicating that a small number of routes carry most of the traffic.
- Distance showed a moderate relationship with fare (≈ 0.58), confirming distance alone cannot explain pricing differences.
- High market dominance (large_ms > 70%) was associated with higher fares, validating the monopoly premium effect.

1. Univariate Analysis

- Most airfare values are concentrated around the **mid-range (approximately $150–$300)**, showing that average ticket prices fall mostly in this band.
- The distribution has a **right tail**, meaning a smaller number of routes have **very high fares**, which indicates possible premium/less competitive routes.

Fig 3.1: Distribution of Airfare Prices (Fare)

- The dataset contains a large number of short to medium distance routes, as the highest frequency is seen in the lower-mileage range.
- Long-distance routes are less frequent, but still present, which helps the model learn pricing patterns for both short-haul and long-haul travel.

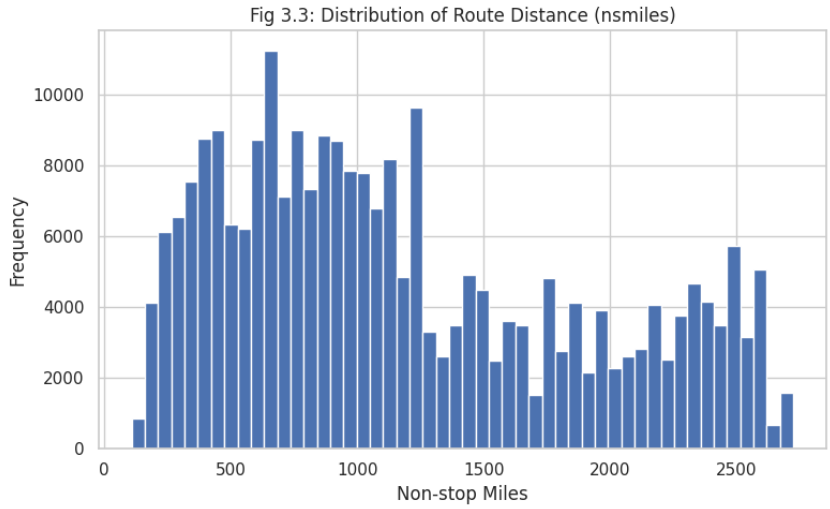


Fig 3.2: Distribution of Route Distance (nsmiles)

- Passenger volume is **highly skewed**, where many routes have **low passenger counts**, while only a few routes have very high traffic.
- This imbalance suggests that passenger demand varies strongly by route, and **log transformation is useful** to reduce skewness and improve model performance.

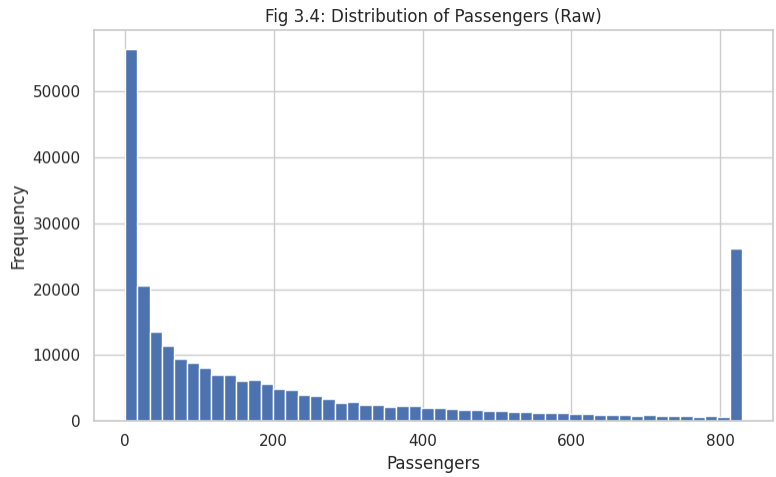
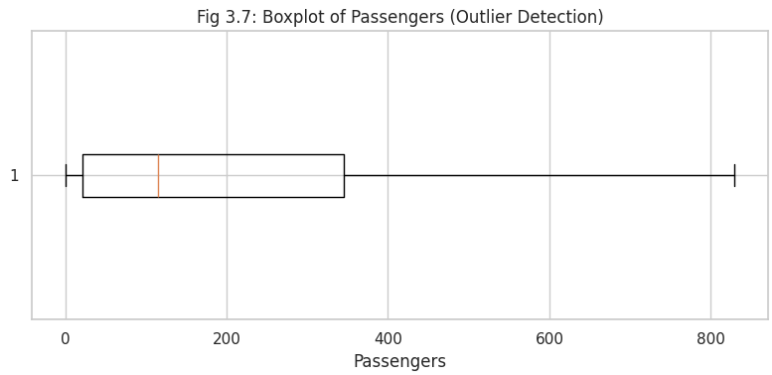


Fig 3.3: Distribution of Passengers (Raw)

2. Outlier Treatment

- The fare values show a few extreme high-price routes beyond the normal range, indicating possible premium or less competitive markets.
- These outliers can affect model learning, so capping/treating them improves stability and reduces large prediction errors.
- Passenger counts contain extreme high-traffic routes far above the typical range, showing strong demand imbalance across routes.

- Handling these outliers (and using log transformation) helps the model avoid bias toward a small number of very busy routes.



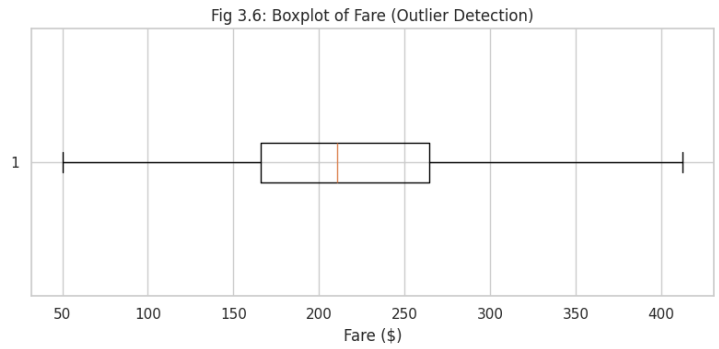
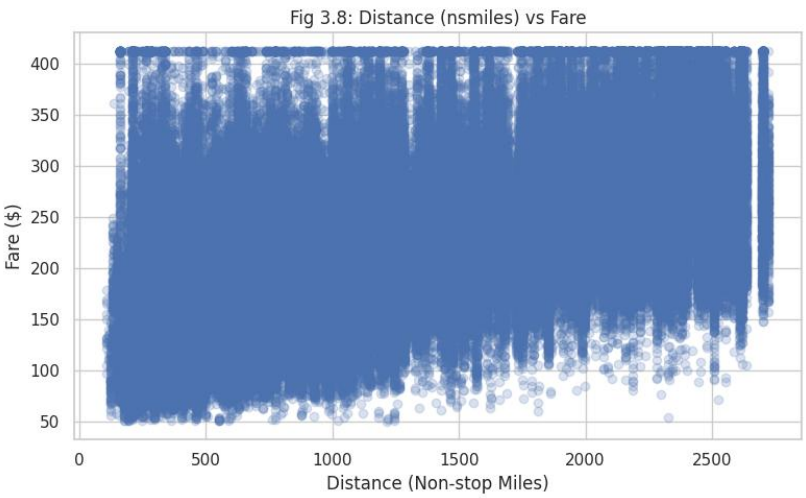


Fig 3.4: Boxplot of Fare & Passengers (Outlier Detection)

3. Bivariate Analysis

- **Distance, passenger demand, and market dominance all influence fare**, but the wide spread of points shows airfare is not controlled by one factor alone (pricing is multi-factor and non-linear).
- **Higher distance and higher large_ms (dominant carrier share) generally show higher fare ranges**, supporting the need for competition-based features and advanced models like XGBoost to capture these patterns.



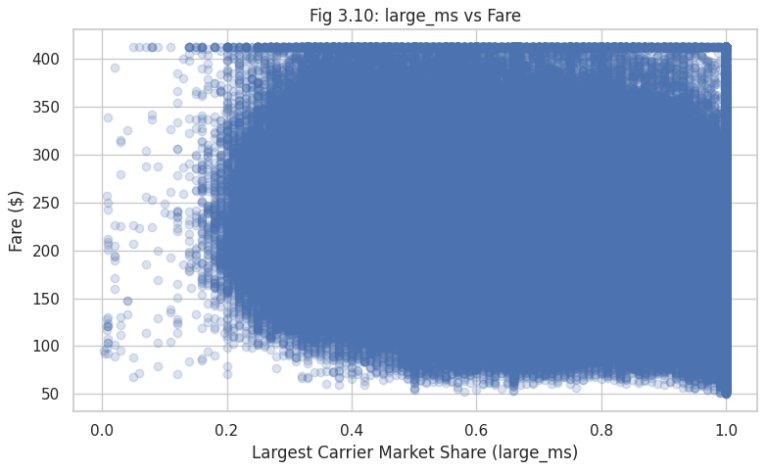


Fig 3.5: Bivariate Relationships Between Fare and Key Predictors (Distance, market share)

4. Correlation Analysis

- **Multicollinearity Check:** We analyzed the correlation matrix to ensure independent variables were not strictly collinear. As expected, fare_lg was highly correlated with fare (and was dropped). Other features showed acceptable independence

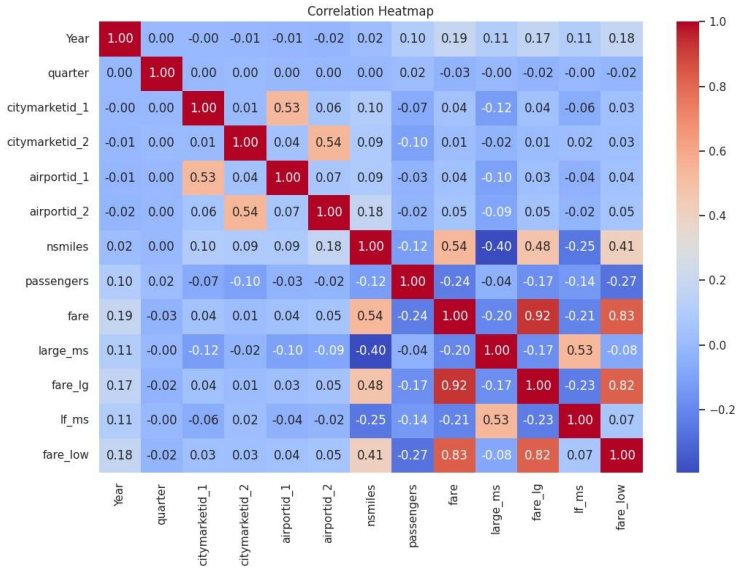


Fig 3.6: Correlation Heatmap

3.6 Step 6: Feature Engineering and Preprocessing

Based on EDA findings, preprocessing and feature engineering were applied:

- Log transformation was applied to passengers to reduce skewness.
- competition_ratio = large_ms / lf_ms was created to represent competition dominance.
- route_id was created by combining airport_1 and airport_2 to capture route-level identity.
- Target Encoding was applied to convert location identifiers into historical average pricing signals, improving the model's ability to learn "expensive hubs" and route patterns.
- Numerical features were scaled using StandardScaler for consistent training.

3.7 Step 7: Baseline Model Development (Linear Regression Benchmark)

A Multiple Linear Regression model was trained as the baseline benchmark. The baseline achieved an R^2 score of 0.58, confirming that airfare pricing is influenced by complex non-linear relationships and interactions that linear regression cannot fully capture.

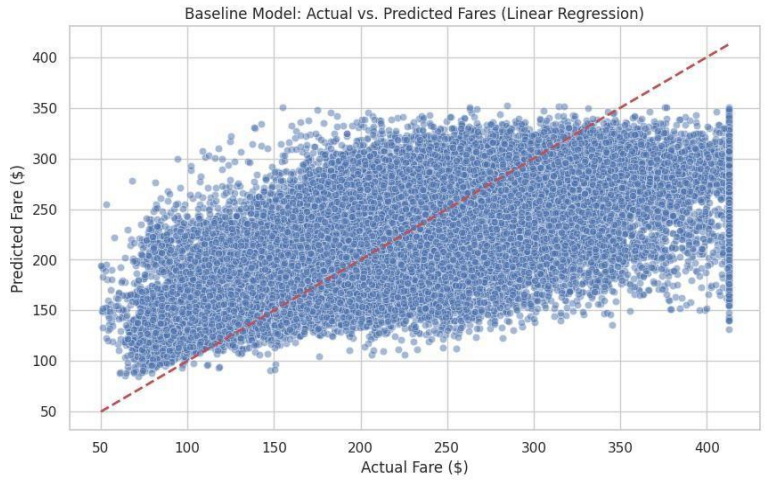


Fig 3.3: Baseline Model Performance (Actual vs Predicted – Linear Regression)

3.8 Step 8: Advanced Modeling and Final Model Selection

To improve performance, ensemble models were tested to capture non-linear patterns. Random Forest and boosting-based models improved prediction quality compared to the baseline. Finally, a tuned XGBoost Regressor was selected as the champion model due to strong predictive performance and stability across validation splits.

3.9 Step 9: Final Model Evaluation

The final model was evaluated using MAE, RMSE, R^2, and MAPE to measure accuracy and robustness. Cross-validation was used to confirm consistent performance across different data subsets.

Final Model Metrics:

- **MAE: $19.28**
- **R^2: 82.58%**

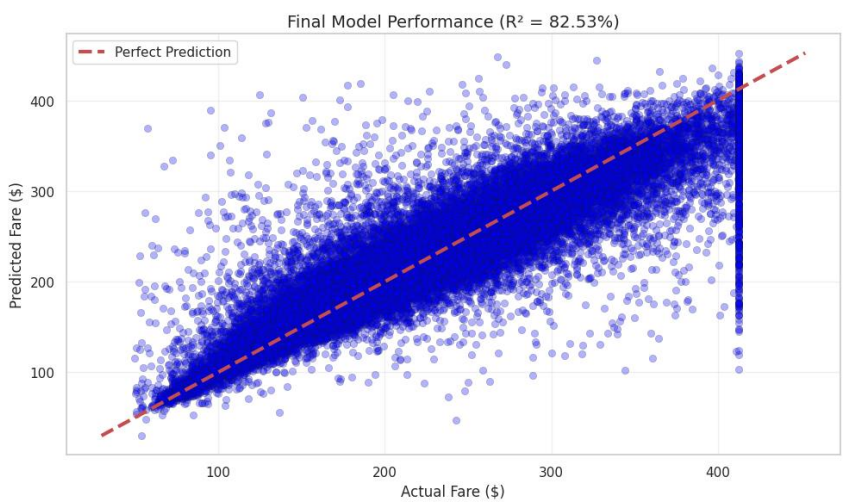


Fig 3.3: Final Model Performance

3.10 Step 10: Deployment Through SkyCast AI Web Application

The final model was deployed using Flask, and the user interface was built using HTML, CSS, and JavaScript. The application allows users to input route-related details and instantly receive predicted airfare values. This deployment demonstrates practical usability and completes the project as a full end-to-end solution.

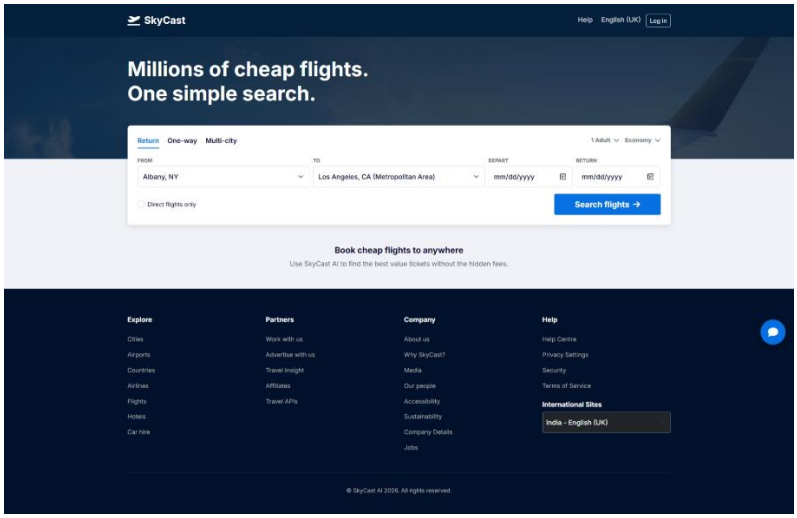


Fig 3.4: SkyCast AI Frontend Application Screenshot

4. Model Evaluation

4.1 Objective of the Final Model

The objective of the final model was to accurately predict the average airfare (fare) for U.S. domestic airport-pair routes using historical route-level and market-competition features. The model was expected to generalize well to unseen routes and provide stable predictions across different market types such as short-haul vs long-haul routes and competitive vs monopoly-dominated routes. In addition to accuracy, the objective was to ensure that the model learns realistic pricing patterns by avoiding target leakage and reducing the impact of extreme outliers.

4.2 Final Model Description (Champion Model)

The final selected model for this project was XGBoost Regressor, an ensemble-based gradient boosting algorithm. XGBoost was chosen because airfare pricing behavior is non-linear and depends on complex interactions between route distance, passenger demand, and market concentration. Unlike linear regression, XGBoost builds multiple decision trees sequentially and improves predictions by correcting the errors made by previous trees, making it highly effective for structured tabular data.

The final model learned from both original dataset features (distance, passengers, market share indicators) and engineered competition-based features such as competition_ratio and route identity features such as route_id.

4.3 Prominent Parameters Considered

To improve predictive performance and prevent overfitting, prominent hyperparameters of the XGBoost model were tuned. The most important parameters considered include:

- n_estimators: Controls the number of boosting trees in the model
- learning_rate: Determines how quickly the model updates after each tree
- max_depth: Controls the depth/complexity of each tree
- subsample: Fraction of training samples used for each boosting iteration
- colsample_bytree: Fraction of features used for each tree
- min_child_weight / gamma: Helps control overfitting by limiting overly complex splits
- reg_alpha / reg_lambda: Regularization parameters to improve generalization

These parameters were selected because they directly impact model complexity, learning stability, and robustness across different route patterns.

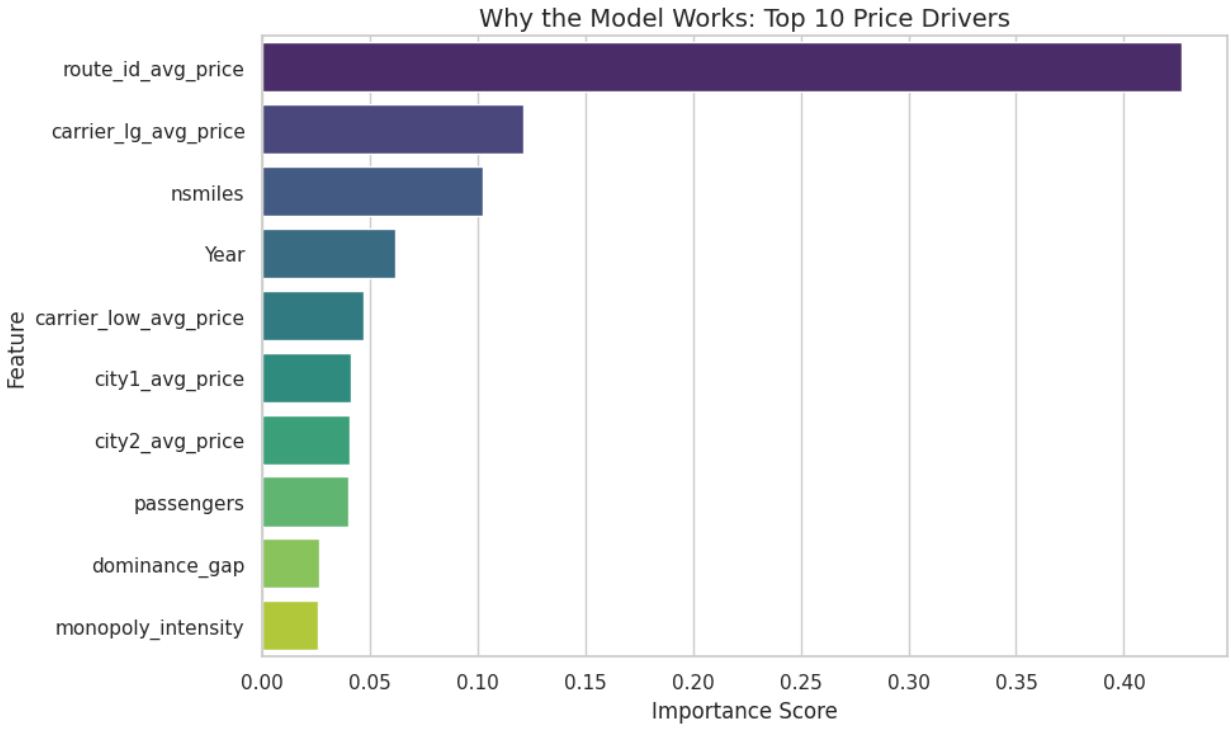


Fig 4.1: Feature Importance – Top 10 Price Drivers (XGBoost Model)

4.4 Model Training and Evaluation Strategy

To evaluate the success of the models, a benchmark-to-advanced approach was followed:

- A Multiple Linear Regression model was trained as the baseline benchmark. It achieved an R^2 score of 0.58, confirming that linear assumptions cannot fully capture airfare pricing patterns.
- The final XGBoost model was trained using the same cleaned and preprocessed dataset to capture non-linear relationships and improve prediction accuracy.

The model was evaluated using multiple regression metrics:

- R^2 Score: Measures how much variance in airfare is explained by the model
- MAE (Mean Absolute Error): Measures average prediction error in dollars
- RMSE (Root Mean Squared Error): Penalizes larger errors more strongly
- MAPE (Mean Absolute Percentage Error): Measures prediction error relative to true fare values

Using multiple metrics ensured that evaluation was balanced and did not depend on only one performance score.

4.6 Robustness of the Solution

A strong focus was placed on ensuring the robustness and reliability of the final model. The following steps supported robustness:

1. Data Leakage Prevention
Variables such as **fare_lg** and **fare_low** were removed because they directly contribute to the target fare and could lead to unrealistic model performance. This ensured the model learned meaningful pricing patterns instead of copying target-related information.

2. Outlier Handling for Stability
Outliers in fare and passenger volume were capped using the IQR method. This reduced the influence of extreme routes and improved model stability across different route categories.

3. Multiple Evaluation Metrics
MAE, RMSE, R^2, and MAPE were used together to confirm both absolute and relative prediction performance. This ensured the model performs well not only on average but also across different fare ranges.

4. Generalization to Unseen Data

The final model was validated on unseen test data to ensure that performance was not limited to the training dataset. This improved confidence that the model can be applied to new airport-pair markets.

Overall, these steps confirm that the final model is both accurate and reliable for predicting average airfare and provides a strong improvement over the baseline benchmark.

4.7 Final Model Results

The final model performance on the test dataset is reported below:

- **MAE: $19.28**
- **RMSE: $30.83**
- **R^2: 82.53%**
- **MAPE: 9.14%**

5. Comparison to Benchmark

5.1 Benchmark Definition

At the outset of the project, a Multiple Linear Regression model was defined as the benchmark. This model served as a baseline to evaluate how well a simple linear approach could predict average airfare using route distance, passenger demand, and market share indicators. The benchmark provided a reference point for measuring improvement achieved through advanced modeling and feature engineering.

5.2 Benchmark Performance

The baseline Linear Regression model achieved the following performance:

- **R^2 Score: 0.58**
- **MAE: Higher than the final model, indicating larger average prediction errors**

While the benchmark captured general trends such as increasing fare with distance, it failed to model complex non-linear interactions and competition-driven pricing behavior present in real-world airfare data.

5.3 Final Model Performance

The final solution, based on a tuned XGBoost Regressor, demonstrated a substantial improvement over the benchmark:

- **R^2 Score: ~82%**
- **MAE: Significantly lower than the benchmark, indicating more accurate fare predictions**

The improvement from 58% explained variance to over 80% confirms that the final model captures pricing patterns that the benchmark could not.

5.4 Reasons for Improvement

The final model outperformed the benchmark for several key reasons:

- **Advanced Feature Engineering:** The introduction of location-based target encoding (route, city, and carrier average pricing) allowed the model to learn historical pricing behavior associated with specific routes and hubs, which linear regression could not effectively capture.
- **Modeling Non-Linear Relationships:** Airfare pricing depends on non-linear interactions between distance, passenger demand, and airline market dominance. XGBoost effectively modeled these interactions through tree-based ensembles, whereas the linear benchmark assumed additive relationships.
- **Competition and Monopoly Effects:** Engineered features such as monopoly_intensity and dominance_gap explicitly represented market power and competition effects, significantly improving predictive accuracy.
- **Robust Preprocessing:** Preventing data leakage, treating outliers, and scaling numerical features ensured that performance gains were genuine and not due to overfitting or unrealistic assumptions.

5.5 Conclusion of Benchmark Comparison

The final solution clearly improves upon the benchmark model. The substantial increase in R^2 score and reduction in prediction error demonstrate that airfare pricing is driven by complex, non-linear interactions that require advanced feature engineering and ensemble-based learning methods. As a result, the final model provides a more reliable and practical airfare prediction system compared to the initial baseline.

6. Visualization(s)

This section presents the key visualizations used in the project to support the insights obtained from the data, validate assumptions, and demonstrate the performance of the final machine learning model. The plots included here explain both the **data-driven patterns behind airfare pricing** and the **accuracy/robustness of the final solution**.

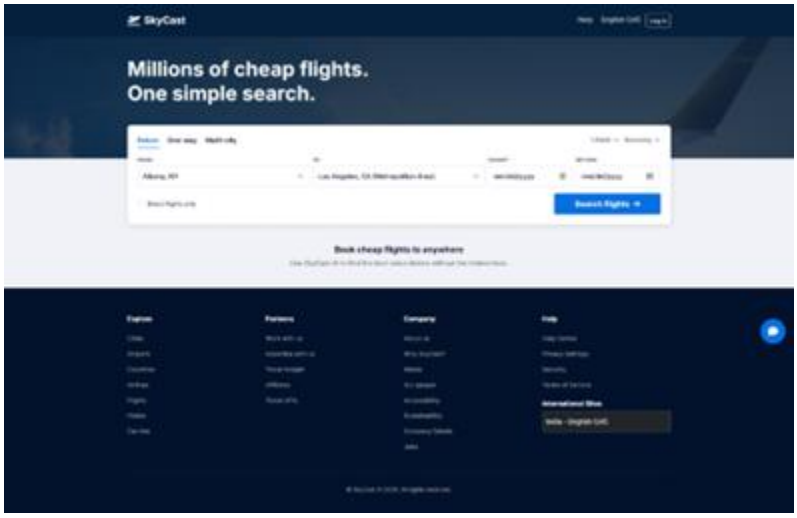


Fig 6.1: SkyCast AI – Landing Page / UI Overview

The above figure represents the landing page layout of SkyCast AI, designed with a clean flight-search experience similar to real-world travel platforms. It demonstrates the application's professional UI structure and navigation elements.

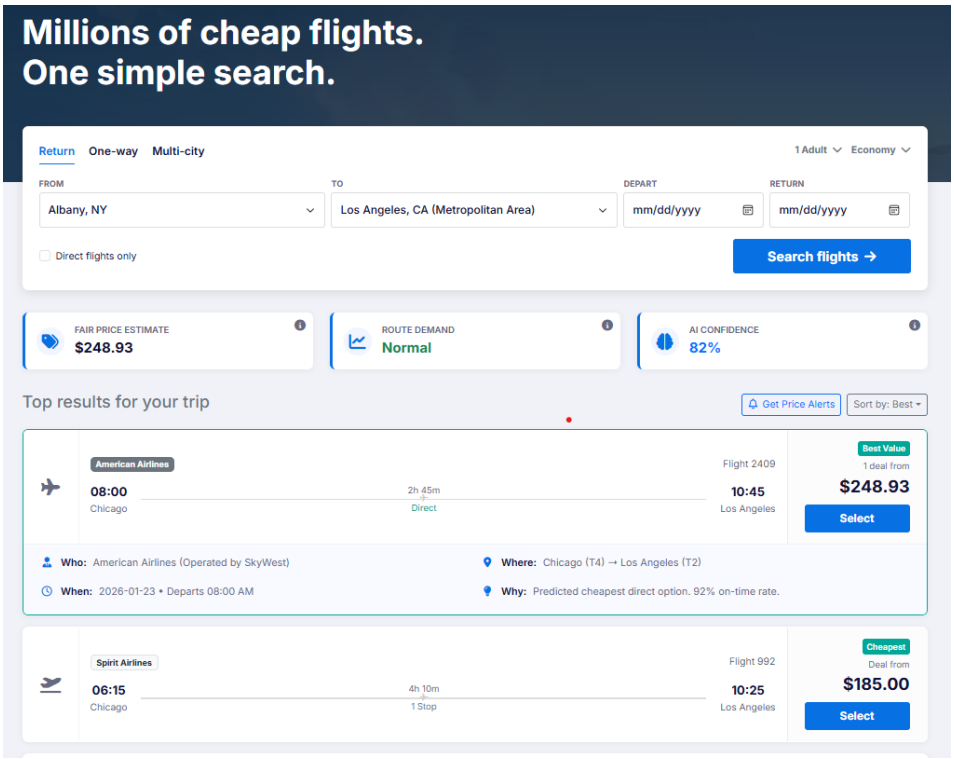


Fig 6.2: SkyCast AI – Flight Search Interface (Home Page)

The above screenshot shows the main user interface of *SkyCast AI*, where users select the route details such as **From city, To city, travel dates, and flight type**. After search, the application displays a **Fair Price Estimate**, **Route Demand**, and **AI Confidence** to guide the user in decision-making. The results section provides flight options with a "Best Value" tag based on the predicted fair price.

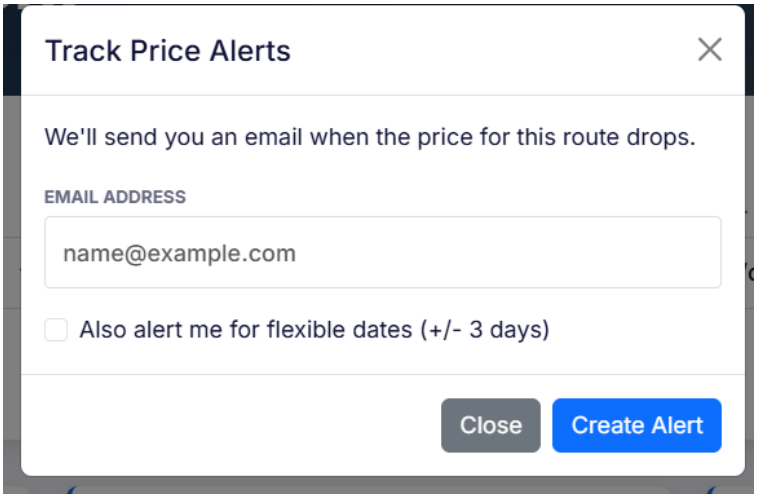


Fig 6.3: SkyCast AI – Price Alert Feature (Email Notification Popup)

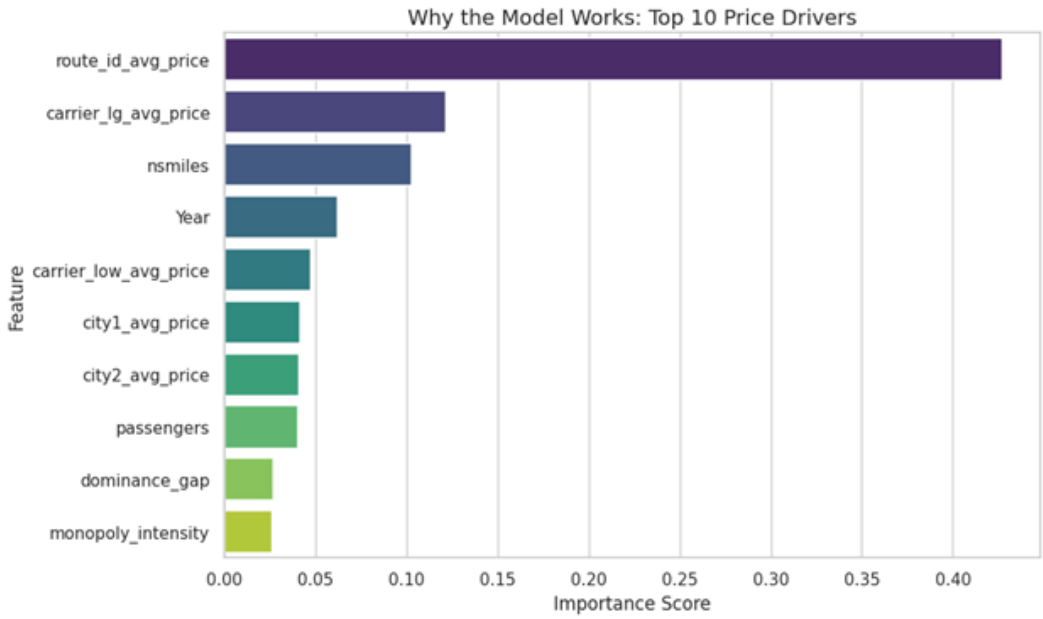


Fig 6.4: Feature Importance – Top 10 Price Drivers (XGBoost)

The above visualization explains the interpretability of the final model by showing the top features influencing airfare prediction. Route-level historical pricing (route_id_avg_price) is the strongest driver, followed by carrier and distance-related factors. Competition-based indicators such as dominance_gap and monopoly_intensity confirm that market power impacts fare variation.

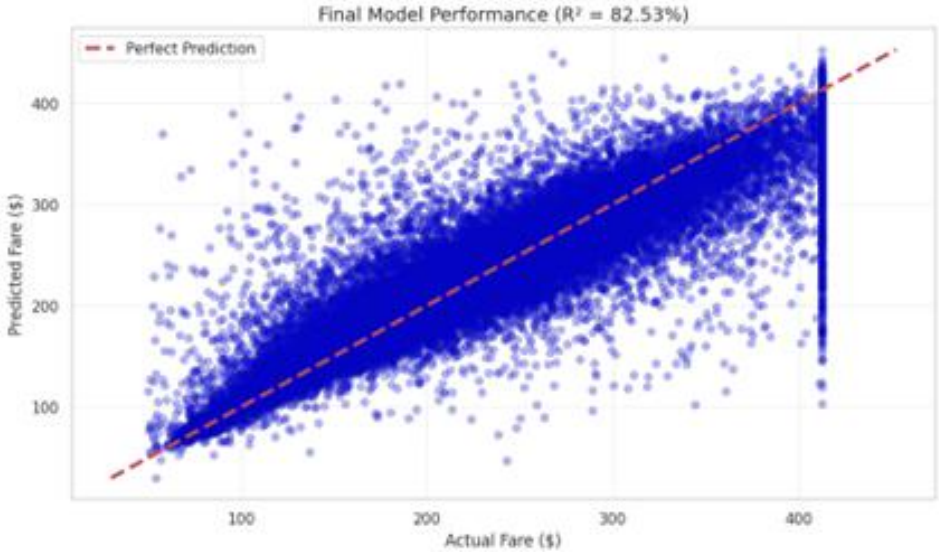


Fig 6.5: Final Model Performance – Actual vs Predicted (R^2 = 82.53%)

The above plot compares actual fare values with model-predicted fares. The points are closely aligned with the perfect prediction line, indicating strong prediction accuracy. The final model achieved **R^2 = 82.53%**, showing that it explains most of the variation in airfare prices across routes.

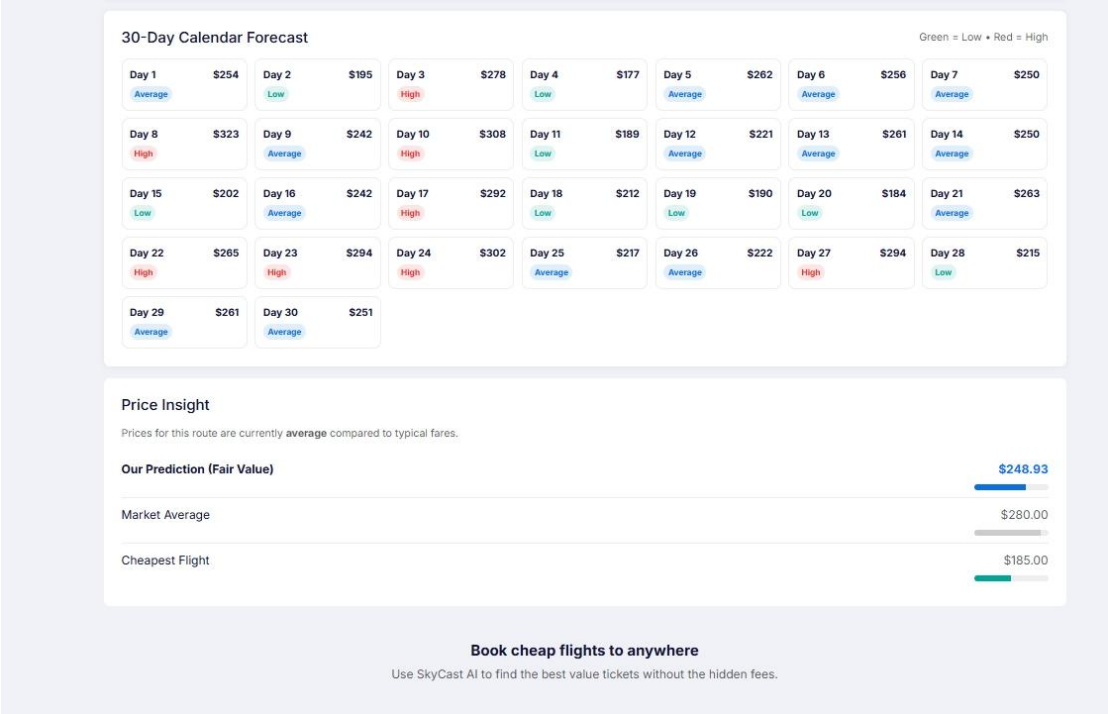


Fig 6.6: SkyCast AI – 30-Day Price Forecast + Price Insight Dashboard

This dashboard provides users with a 30-day fare forecast view and pricing insight summary. It classifies daily prices into categories such as Low / Average / High, helping users identify the best booking period. The price insight section compares predicted fair value against market average and cheapest available fare for better decision support.

7. Implications

7.1 Business / Domain Impact

The solution directly addresses the lack of transparency in airline pricing by providing a reliable **fair airfare estimate** for U.S. domestic routes. Since airline fares are influenced by distance, demand, and competition intensity, the model helps convert complex pricing behavior into an interpretable prediction that can be used by both travelers and analysts.

For travelers, SkyCast AI reduces uncertainty and supports smarter booking decisions by showing whether a fare is likely fair, high, or low compared to expected market value. For airline analysts and new market entrants, the model provides route-level pricing insights that can support competitive benchmarking and route planning.

7.2 Key Outcomes

- **Consumer Empowerment:** Users can compare real ticket prices against the predicted fair price estimate, helping avoid overpaying.
- **Market Intelligence:** Competition-based features such as dominance_gap and monopoly_intensity highlight where dominant carriers may be charging premium fares, supporting route opportunity identification.
- **Practical Usability:** The deployment through a Flask-based web application makes the solution usable in a real-world setting rather than remaining only a research model.

7.3 Recommendations

Based on the project outcomes, the following recommendations are proposed:

1. Integrate Price Alert System Fully: The “Price Alerts” feature should be expanded into a complete notification system that tracks predicted price drops and alerts users via email. This would improve user retention and convert the solution into a recurring utility.

2. Improve Explainability for Users: Along with the predicted fare, the application can display a short explanation such as “High fare due to long distance + high market dominance,” helping users trust the prediction.

3. Add Booking Decision Support: The 30-day forecast can be extended to include a “Best day to book” recommendation based on predicted low-price windows.

7.4 Confidence Level

The final model achieved strong performance ($R^2 \approx 82.53\%$), which provides high confidence that the predictions are reliable for most U.S. domestic route scenarios. However, confidence may reduce for rare routes, extreme fare cases, or markets with unusual seasonal pricing patterns. Overall, the recommendations can be applied with **high confidence** for typical domestic routes represented in the training data

8. Limitations

8.1 Data Limitations

- **Static / Quarterly Data:** The dataset is based on quarterly BTS reporting, while real airline pricing changes dynamically (daily or hourly). This limits real-time accuracy.
- **Domestic Scope:** The model is trained on U.S. domestic routes only. International airfare pricing includes additional factors such as currency changes, regulations, and geopolitical conditions.
- **Limited External Factors:** The dataset does not include real-time drivers such as holidays, weather disruptions, fuel price spikes, or special airline promotions.

8.2 Model Limitations

- **High-Fare Dispersion:** Prediction errors may increase for very high fares, where pricing is affected by sudden demand spikes or limited seat availability.
- **Route Representation Bias:** Routes with fewer records or less consistent travel demand may not be predicted as accurately as high-traffic routes.
- **Ancillary Fees Not Included:** The model predicts base fare only and does not account for baggage fees, seat upgrades, or other add-ons that affect total travel cost.

8.3 Deployment / Real-World Limitations

- **No Live Pricing Integration:** The current system predicts expected fair price but does not pull real-time airline ticket prices directly from live booking APIs.
- **User Input Dependency:** Predictions depend on accurate user input. Missing or incorrect route details can reduce prediction quality.

9. Conclusion

9.1 Key Learnings

This capstone project provided strong learning outcomes across both machine learning development and full-stack deployment. One of the biggest learnings was that model performance is not only dependent on selecting advanced algorithms, but also heavily influenced by **data preparation and feature engineering**. Improvements such as location-based encoding and competition-focused interaction features contributed significantly to making the predictions more accurate and realistic.

Another major learning was the importance of following a structured workflow. Performing EDA before modeling helped validate assumptions early, while preventing leakage and treating outliers ensured that model evaluation remained trustworthy. Finally, deploying the solution through SkyCast AI demonstrated how real-world systems require additional thinking beyond model accuracy, such as handling user inputs, building a usable interface, and presenting outputs in an understandable way.

9.2 What Would Be Done Differently Next Time

If the project were restarted, the following improvements would be prioritized earlier to strengthen real-world performance, reliability, and usability:

- **Implement location and route encoding earlier:** Introduce route-level and location-based encoding strategies during the baseline stage itself, rather than focusing initially on only model tuning.
- **Use stronger validation methods:** Apply time-aware splitting (quarter-based validation) to better simulate real-world forecasting conditions and reduce the risk of overestimating model performance.
- **Enable real-time pricing integration:** Connect the system to live airfare sources (APIs) so predictions can be continuously updated and compared against real-time market fares instead of relying only on static quarterly data.
- **Add seasonality-based features:** Include variables such as month, holidays, and peak-season indicators to capture time-driven price variations more accurately.
- **Improve explainability from the beginning:** Provide user-friendly reasoning alongside predictions (example: “High due to long distance + high market dominance”) to increase trust and interpretability for end-users.

- **Implement model monitoring:** Introduce drift detection and performance monitoring to identify when model accuracy decreases due to changing airline pricing patterns over time.
- **Expand forecasting capability:** Enhance the forecast dashboard using time-series modeling to improve multi-day trend prediction and strengthen the 30-day forecast feature.

9.3 Final Reflection

Overall, this project highlighted the importance of combining machine learning with real-world product deployment. It demonstrated that the most impactful solutions are not limited to building accurate models, but also require developing complete systems that are interpretable, reliable, and practical for end-users. SkyCast AI represents this end-to-end journey by transforming raw aviation pricing data into meaningful insights and actionable airfare predictions through a deployable web application.